

Multidimensional Transport Poverty Index

A replicable multi-dimensional and multi-modal measure of transport poverty

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Abstract

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Keywords: keyword1, keyword2

1. Introduction

Transportation is an essential service which can condition access to almost all human activities, be them work, education, health or social life. When this access is limited by insufficient supply, high costs, or elevated exposure to systemic risks and externalities, the term Transport Poverty is used (Lucas et al., 2016). This phenomenon has direct

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consequences in social exclusion and territorial cohesion ([Mejia-Dorantes and Murauskaite-Bull, 2022](#)).

In the European context, this problem has increased its political relevance over the last decade. The European Pillar of Social Rights, in its 20th Principle, enshrines the right to access essential services, including transportation. Regulation (EU) 2023/955 of 10 May 2023, which established the Social Climate Fund (SCF), enshrined the first legally bounding definition of transport poverty at an European scale, defining it as “individuals’ and households’ inability or difficulty to meet the costs of private or public transport, or their lack of or limited access to transport needed for their access to essential socioeconomic services and activities, taking into account the national and spatial context” ([Parliament and the Council of the European Union, 2023](#)) .

Despite the clear empirical relevance of this phenomenon, there is difficulty in quantifying it in a systematic way. Specifically in Portugal, previous research has focused on punctual case studies ([Pritchard et al., 2014](#)) or on costly mobility surveys ([INE, 2018](#)), which can hardly be renewed at the cadence required by public policy cycles.

This project was thus developed in an attempt to bridge the existing methodological gap with the creation of a composite and intermodal tool - the Multidimensional Transport Poverty Index (IMPT) - that is replicable and scalable, as well as wholly based in public data, intended to assist and support public policy instruments, such as the European Union’s Social Climate Fund (SCF) or Sustainable Urban Mobility Plans (SUMP). The Lisbon Metropolitan Area (LMA) was employed as a case study for the tool’s validation.

2. Literature review

2.1. Definitions of Transport Poverty

Transport systems are a fundamental part of modern society, and their absence is related to social exclusion - a state where an individual or a societal group is left out of participating in regular social activities in a given society ([Pritchard et al., 2014](#)) -, socio-territorial inequality and transport inequality - the differences in transport conditions for different groups of people ([Hidayati et al., 2021](#)). Despite the relation between poverty and

transport conditions, these are also related to a wide range of social issues. As an example, motorized vehicle users contribute the most to externalities, but vulnerable road users are disproportionately affected by them in health, space allocation and time consumption (Gössling, 2016). Moreover, lower-income households tend to not have financial conditions to have motorized transport or to reside in central areas, relegating them to live in areas distant from the city center, often poorly served by public transportation. (Fedorowicz et al., 2020) (Lopes et al., 2023).

These differences in transport conditions can lead to transport poverty, a concept which lacks a single universally accepted definition in scientific literature. This expression has been used to broadly describe inequities related with transportation access, but also in a more restricted way to refer to the economic inability to support transportation costs (Mattioli et al., 2018). However, as Lucas et al. (2016) states, different terminologies relating to transport poverty - terms such as “mobility poverty”, “accessibility poverty” or “transport disadvantage” - are used with “often very different, although also overlapping definitions”. Due to this lack of a universal definition and the multidimensional nature of transport poverty, its measurement and quantification poses a challenge (Levinson and King, 2020).

This way, the necessity to comprehend this concept and quantify it becomes more urgent. It is essential to define reliable indicators of transport vulnerability to map “public transport deserts” (Jiao and Dillivan, 2013) and low-accessibility areas in order to inform public intervention and to monitor the implementation of measures to improve these situations. A reliable and replicable index to measure transport poverty is thus fundamental to support public policy instruments.

Lucas et al. (2016) proposes the most comprehensive conceptual systematization of transport poverty, clearly defining previously different but overlapping terminology. They define transport poverty as a combination of the following four dimensions:

- Accessibility - the capacity to reach usual destinations and activities (such as work, education or health).
- Mobility - the systemic aspects of the transportation system itself that impact

essential trips, expressed in parameters such as public transport service frequency or average commuting time.

- Affordability - the ability of households to meet the cost of transportation.
- Exposure to externalities (Safety) - the negative effects of exposure to the transport system itself, such as environmental pollution or road crash rates.

Recently, and based on the aforementioned concepts, [Mejia-Dorantes and Murauskaite-Bull \(2022\)](#) have proposed a definition that has been widely adopted: transport poverty is the inability of an individual or household to access essential services or opportunities due to transportation services that are inaccessible, inadequate, unaffordable, time-consuming or unsafe. This definition has been adapted and applied at the European scale, through [Parliament and the Council of the European Union \(2023\)](#) and a recent European Commission Report ([European Commission. Directorate General for Employment and Inclusion, 2024](#)). The key dimensions in the EC Report are synthesized as *Availability* (corresponding to Lucas' Mobility), *Accessibility* and *Affordability*, with exposure to externalities being included in the overarching concept of *Adequacy* (Figure 1).

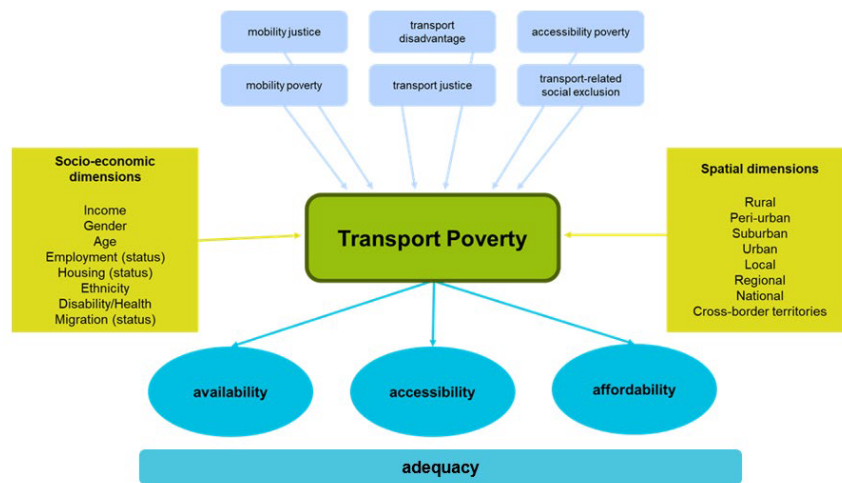


Figure 1: Conceptualization of Transport Poverty. Source: Oeko-Institut and University of Manchester

In summary, the concept of transport poverty has evolved from a one-dimensional look at accessibility or affordability, to a composite approach that attempts to fully capture

the multidimensional nature of this phenomenon.

2.2. Dimensions of Transport Poverty

The dimensions proposed by [Lucas et al. \(2016\)](#) continue to be the reference for composite indexes of transport poverty ([Alonso-Epelde et al., 2023](#)) ([Mejia-Dorantes and Murauskaite-Bull, 2022](#)). IMPT adopts these dimensions, implementing them through secondary data available in Portugal. Each dimension is further explained thoroughly.

2.2.1. Accessibility

Accessibility refers to the ability to reach regular destinations such as employment, education, health and other essential services ([Allen and Farber, 2019](#)) ([Unit, 2003](#)). The literature distinguishes between approaches based on cumulative opportunities - the number of reachable destinations within a certain travel time threshold -, gravity-based approaches, which continuously weight destinations based on impedance (i.e. the easier a destination is to reach, the more weight it will have) and hybrid approaches ([Geurs and van Wee, 2004](#)) ([Klar et al., 2023](#)). [Pereira and Herszenhut \(2023\)](#) formalize these methodologies in open-source software *R*, through the library *r5r*, which provides the technical base for the Accessibility dimension of the IMPT ([Pereira et al., 2021](#)).

2.2.2. Mobility

Mobility refers to the systemic deficits of the transport system, be them lack of infrastructure, lack of service frequency or long trip duration ([Levinson and King, 2020](#)) ([Lucas et al., 2016](#)) ([Mejia-Dorantes and Murauskaite-Bull, 2022](#)). This dimension captures different aspects of transport supply that accessibility is not able to, their core difference being that, while accessibility asks which places are reachable, mobility focuses on how easy it is to reach them.

2.2.3. Affordability

This dimension measures the capacity of households to support essential transportation expenses ([Isalou et al., 2014](#)) ([Litman, 2026](#)) ([Oviedo et al., 2025](#)). The literature distinguishes three groups of indicators:

- The *10 percent rule*, which considers the households who spend more than 10% of their income in transportation, having been adopted in the United Kingdom, with its origins in the concept of “fuel poverty” (Lucas et al., 2016) (Alonso-Epelde et al., 2023).
- The *2M* approach, adapted from the concept of energy poverty, which considers the households that spend in transportation more than twice the national median (Alonso-Epelde et al., 2023).
- The combined approach of *housing and transportation affordability*, based on the premise that households make their residential location choice based on both housing and transportation costs. If the combined housing and transport cost doesn’t exceed 45% of income, it is considered affordable (Isalou et al., 2014) (Litman, 2026).

The IMPT was rooted upon this last indicator, combining the costs of housing and transportation in relation to household income.

2.2.4. *Safety*

This dimension is commonly not included in transport poverty indexes (Kelly et al., 2023), being considered only in the most wide-ranging definitions of transport poverty (Lucas et al., 2016) (UN-Habitat, 2013). It refers to the exposure to externalities of the transport system, such as road crashes and pollution, which have disproportionate effects on the territory. For the IMPT, this dimension’s inputs only included those related to road safety indicators, due to the inability to get open and granular data regarding environmental externalities.

A fifth dimension, digital literacy, was initially considered as a proxy for access to information and mobility services (Durand et al., 2022) (Orozco-Fontalvo et al., 2025) (Velaga et al., 2012). However, it was excluded from the final implementation of the index due to unavailability of data compatible with the scale of the lowest administrative division.

2.3. Examples of geospatial tools of accessibility and equity

During the last decade various interactive spatial tools based on indicators of accessibility and inequality have been created and made publicly available. Two of these are here highlighted, due to their conceptual or methodological influence and their proximity to the IMPT's objective.

The project *Acesso a Oportunidades* (Access to Opportunities), developed by Instituto de Pesquisa Econômica (IPEA) under Rafael H. M. Pereira, provides accessibility estimates for the twenty biggest cities in Brazil, disaggregated in a hexagonal grid, by transport mode and type of opportunity. Its contribution is the creation of open R code, with the packages `r5r` and `accessibility`, which made the computation of multimodal travel time matrices possible (Pereira et al., 2021) (Pereira and Herszenhut, 2022) (Pereira and Herszenhut, 2023). An interactive tool to visualize these results was also created, and is available at: <https://www.ipea.gov.br/acessoootportunidades/mapa/>. IMPT directly incorporates this technical infrastructure in its Accessibility dimension, in order to compute the number of reachable opportunities by different travel modes and travel time thresholds, for each grid cell in the study area. Pereira's work is this way a critical operational pillar of the IMPT.

Another example is the *Atlas of Inequality*, developed by MIT Media Lab and Northwestern University, under the coordination of Esteban Moro. This project is originated from a different problem, and attempts to measure *segregation by experience*, i.e. how people of different socioeconomic levels share the same urban spaces in daily life, through anonymized cell phone location data. This indicator thus captures inequality in the use of the city, and not transport supply or access to destinations. It is available at: <https://inequality.media.mit.edu>.

However, the methodological distinction between *Acesso a Oportunidades*, *Atlas of Inequality* and the IMPT is significant. *Acesso a Oportunidades* is a tool for the measurement of accessibility, i.e. what is reachable from each location, under assumptions on transport supply, but not taking into account the quality of the transport service offered, the economic capacity of households or exposure to externalities. The *Atlas of Inequality*, is a tool to measure segregation, which is often revealed by mobility patterns.

It captures the social dimension of urban inequality without diagnosing transport poverty itself, due to the exclusion of other variables of transport supply, exposure to externalities and costs. IMPT, on the other hand, is a multidimensional index to measure transport poverty, aggregating the dimensions of Accessibility, Mobility, Affordability and Safety in a composite index, in line with the operational definition provided by the European Union ([Parliament and the Council of the European Union, 2023](#)) ([European Commission. Directorate General for Employment and Inclusion, 2024](#)) and supported by the literature ([Lucas et al., 2016](#)) ([Mejia-Dorantes and Murauskaite-Bull, 2022](#)).

2.4. IMPT's Contribution

IMPT emerges in an attempt to create an index synthesizing these dimensions and providing a reliable measurement of transport poverty, bridging three gaps in the literature and in national public policy.

1. There is, in Portugal, the absence of an index that integrates the four aforementioned dimensions across multiple transport modes and is replicable at the lowest administrative level.
2. Existing indexes excessively depend on costly primary data that is difficult to monitor continuously. IMPT relies exclusively on secondary and open data, easily allowing for updates and monitoring over time.
3. There is a lack of operational tools that allow to translate academic literature on transport poverty into decision-support instruments for local governments, transport authorities, and generally public policy makers.

Furthermore, the IMPT incorporates four methodological variants, corresponding to different analytical positions:

- Contrasting IMPT: based on entropy, it highlights the variables with the highest variation, better capturing territorial contrasts.
- Critical IMPT: based on the geometric mean, it non-linearly penalizes the dimension with the worst performance in each geographical unit.
- Balanced IMPT: based on the arithmetic mean, it allows for compensation between dimensions.

- Political IMPT: allowing for a personalized Analytic Hierarchical Process (AHP) weighting, it can be tailored to the decision-maker’s preferences and priorities.

3. Methodology

Based on the guidelines provided by the OECD Handbook (OECD, 2008), the IMPT was computed through the following procedural steps: (1) data collection, computation and normalization of every indicator; (2) aggregation of indicators per dimension through a Principal Component Analysis (PCA), and normalization of each dimension; (3) multi-dimensional aggregation of the four dimensions, employing three distinct methods; (4) results validation, through a workshop with representatives of municipalities within the Lisbon Metropolitan Area (LMA).

3.1. Case Study

The Lisbon Metropolitan Area (LMA) was applied as the case study for the IMPT. The LMA is comprised of two NUTS II regions (Grande Lisboa and Península de Setúbal), which encompasses 18 Municipalities and approximately 2,8 million inhabitants. This region presents territorial characteristics which make it particularly suitable for this analysis: a polycentric structure with a strong employment center in Lisbon, significant discrepancies between the public transport offer in the urban core and in the periphery, and a high modal share of private vehicles in a large part of the metropolitan area.

The analysis is focused on the lowest administrative division, the *freguesia* (parish), which is the most disaggregated geographical level that guarantees data coverage for all utilized sources. For the indicators that had a higher level of granularity, an analysis by grid level was also assembled. The parish results were also aggregated by *município* (municipality), the administrative level immediately above the parish. The LMA contains 18 municipalities and 124 parishes (Figure 2), which constitute the observation units of the IMPT.

3.2. Data Sources

Based on the literature, a preliminary list of indicators for each dimension was compiled, followed by an extensive search for public data to obtain the most possible of these

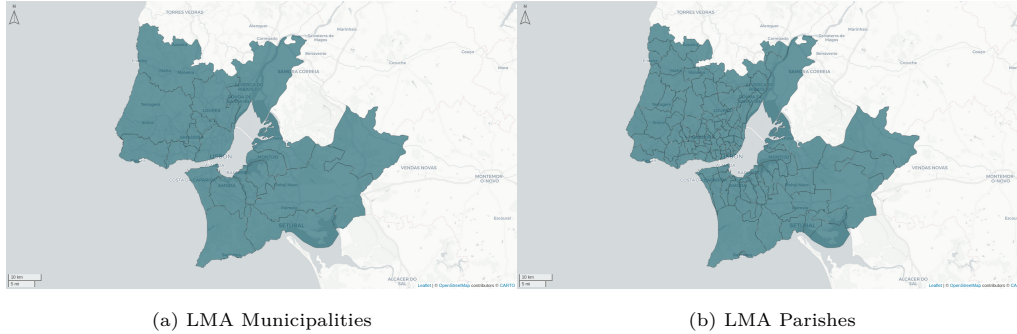


Figure 2: Geographical units of the LMA

indicators. However, due to some of them not being found or not being publicly accessible, the final list of indicators included is shown in Table 1. This excludes indicators such as exposure to air pollution, overcrowding of public transportation, or walkability, as well as the dimension of digital literacy.

```
all_indicators <- data.frame(
  Dimension = c("Accessibility",
    "Accessibility",
    "Mobility",
    "Mobility",
    "Mobility",
    "Mobility",
    "Mobility",
    "Mobility",
    "Affordability",
    "Affordability",
    "Safety"
  ),
  Indicator = c("Number of opportunities within travel time threshold",
    "Travel time to first n opportunities",
    "Average commuting time",
    "Average number of commuting transfers",
```

```

        "Public transport headways and waiting times",
        "Public transport population coverage",
        "Ratio of pedestrian and cycling infrastructure",
        "Shared mobility availability",
        "Transportation costs",
        "Housing and transportation costs",
        "Aggregation of road safety indicators"
    )
)
knitr::kable(all_indicators)

```

Warning in attr(x, "align"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Warning in attr(x, "format"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Table 1: Indicators used in the IMPT

Dimension	Indicator
Accessibility	Number of opportunities within travel time threshold
Accessibility	Travel time to first n opportunities
Mobility	Average commuting time
Mobility	Average number of commuting transfers
Mobility	Public transport headways and waiting times
Mobility	Public transport population coverage
Mobility	Ratio of pedestrian and cycling infrastructure
Mobility	Shared mobility availability
Affordability	Transportation costs
Affordability	Housing and transportation costs

Table 1: Indicators used in the IMPT

Dimension	Indicator
Safety	Aggregation of road safety indicators

From here,

```
data_sources <- data.frame(
  Code = c("CAOP", "BGRI21", "IMOB", "GTFS", "OSM-RN", "OSM-POI", "CM",
           "ASNR", "INE-Inc", "INE-Hous", "GBA", "PMUS", "COS", "IP"),
  Database = c("CAOP 2024.1 - Administrative Boundaries",
               "Census 2021 - Base Reference Geographical Information",
               "IMOB 2017/18 - Inquérito à Mobilidade nas Áreas Metropolitanas",
               "GTFS - General Transit Feed Specification",
               "OpenStreetMap (OSM) - Road network",
               "OpenStreetMap (OSM) - Points of interest",
               "Carris Metropolitana - Health centres and schools datasets",
               "ANSR - Road accident data (location and type)",
               "INE - Household income by parish",
               "INE - Housing costs by parish",
               "Global Building Atlas - Building heights",
               "TML PMUS - Shared mobility docks",
               "COS 2023 v1 - Land use map",
               "Infraestruturas de Portugal - Toll costs"
            ),
  Year = c("2024", "2021", "2018", "2026-02", "2026-01", "2024-04", "2025",
           "2019-2023", "2023", "2021", "2026", "2025", "2023", "2026")
)
knitr::kable(data_sources)
```

Warning in attr(x, "align"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See `help("Deprecated")`

Warning in `attr(x, "format")`: `'xfun::attr()'` is deprecated.

Use `'xfun::attr2()'` instead.

See `help("Deprecated")`

Table 2: Data sources

Code	Database	Year
CAOP	CAOP 2024.1 - Administrative Boundaries	2024
BGRI21	Census 2021 - Base Reference Geographical Information	2021
IMOB	IMOB 2017/18 - Inquérito à Mobilidade nas Áreas Metropolitanas	2018
GTFS	GTFS — General Transit Feed Specification	2026-02
OSM-RN	OpenStreetMap (OSM) — Road network	2026-01
OSM-POI	OpenStreetMap (OSM) — Points of interest	2024-04
CM	Carris Metropolitana — Health centres and schools datasets	2025
ASNR	ANSR — Road accident data (location and type)	2019-2023
INE-Inc	INE — Household income by parish	2023
INE-Hous	INE — Housing costs by parish	2021
GBA	Global Building Atlas - Building heights	2026
PMUS	TML PMUS - Shared mobility docks	2025
COS	COS 2023 v1 — Land use map	2023
IP	Infraestruturas de Portugal — Toll costs	2026

Lastly, the following R packages were utilized in the data processing and creation of both the index and its interactive dashboard: `readr`, `readxl`, `openxlsx`, `lubridate`, `dplyr`, `tidyr`, `stringr`, `tidyverse`, `sf`, `terra`, `mapview`, `stplanr`, `h3jsr`, `r5r`, `accessibility`, `odjitter`, `tidytransit`, `GTFSshift`, `FactoMineR`,

assertthat, piggyback.

3.3. Each Dimension's Indicators

The IMPT is not only multidimensional, but also intermodal. In Table 3, the dimensions used to compute the index for each mode are specified.

```
modes_per_dimension <- data.frame(
  Dimension = c("Accessibility",
               "Mobility",
               "Affordability",
               "Safety"
               ),
  Mode = c("Driving, Public Transport, Walking, Cycling",
           "Driving, Public Transport, Walking, Cycling",
           "Driving, Public Transport",
           "Driving, Walking, Cycling"
           )
)
knitr::kable(modes_per_dimension)
```

Warning in attr(x, "align"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Warning in attr(x, "format"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Table 3: Modes evaluated in each dimension

Dimension	Mode
Accessibility	Driving, Public Transport, Walking, Cycling
Mobility	Driving, Public Transport, Walking, Cycling

Table 3: Modes evaluated in each dimension

Dimension	Mode
Affordability	Driving, Public Transport
Safety	Driving, Walking, Cycling

3.3.1. Accessibility

The Accessibility dimension quantifies the capacity to reach key destinations within a given travel time threshold. The key destination categories, as well as their data sources, are summarized in table (...)

This dimension incorporates two metrics:

1. Cumulative number of opportunities: number of points of interest of a given category reachable within a certain amount of time.
2. Travel time to reach nearest opportunities: time required to reach the first n opportunities of a given category.

Indicators for this dimension were computed using the H3 grid ([Uber Technologies, 2018](#)) of resolution 8 ([Pereira and Herszenhut, 2022](#)), which corresponds to 3686 hexagons in the LMA, with an edge length of 531 meters and an average hexagon area of approximately 737,327m².

3.3.1.1. Jittering.

Due to a lack of open access to fine data regarding job location, the most granular data obtained was an OD matrix at the parish level, provided by [INE \(2018\)](#), with centroid to centroid representations, “a common approach involving the simplifying assumption that all trip destinations and origins can be represented by (sometimes population weighted or aggregated) zone centroids” ([Lovelace et al., 2022](#)). However, in order to accurately compute accessibility and mobility measures, there was the need to have more disaggregated data regarding job locations, beyond the parish unit.

This way, jittering (Lovelace et al., 2022) was used as a proxy for this purpose, using building volumes as trip attractors, and this way weighting origins and destinations. This allowed for the disaggregation of trips between parishes to ODs with up to 50 trips.

3.3.1.2. Travel time matrix.

A travel time matrix was used as the basis for all accessibility measures. It was computed between each grid cell, over a network (Pereira et al., 2021) that was built considering:

- OpenStreetMap road network, exported from HOT (2026).
- General Transit Feed Specification (GTFS) for all operators in the LMA (valid at 04/02/2026).
- Terrain elevation profile (EU, 2026).

The matrix was computed using the function `r5r::travel_time_matrix()` with the following parameters: ()

3.3.1.3. Cumulative number of opportunities.

The cumulative number of opportunities was calculated using the function `accessibility::cumulative_cutoff()` considering the travel time matrices for each grid cell combination, the number of opportunities per grid cell, and the travel time cut-offs.

This measure was computed for all combinations of POIs and modes for time cut-offs of 5, 10, 15, 30, 45, 60, 75 and 90 minutes. However, as can be seen in Table 4, only selected combinations of modes/points of interest were included in IMPT's operationalization.

```
accessibility_n_opportunities <- data.frame(
  "Point of Interest/Mode" = c("Healthcare Facilities",
                              "Basic Education",
                              "Green Spaces",
                              "Recreation",
                              "Groceries",
```

```

        "Jobs"
      ),
      "Driving or Public Transport" = c("30","30","5-10","5-10","15","30-45"),
      "Walking or Cycling" = c("15","15","15-30","15-30","15","30-45"),
      check.names = FALSE
    )
knitr::kable(accessibility_n_opportunities)

```

Warning in attr(x, "align"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Warning in attr(x, "format"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Table 4: Travel time threshold (minutes) to compute cumulative number of opportunities

Point of Interest/Mode	Driving or Public Transport	Walking or Cycling
Healthcare Facilities	30	15
Basic Education	30	15
Green Spaces	5-10	15-30
Recreation	5-10	15-30
Groceries	15	15
Jobs	30-45	30-45

The value computed represents the number of opportunities reachable within the defined travel time threshold. Values at parish and municipality level are aggregated considering a weighted mean of the number of opportunities reachable in each grid cell within that geographical unit, weighted by the population of those cells.

3.3.1.4. Travel time to reach closest opportunities.

This measure was computed using the function `accessibility::cost_to_closest()`, considering the travel time matrices for each grid cell combination and the number of opportunities in each grid cell. Its output corresponds to the travel cost, in minutes, to reach n opportunities of a certain category. Although calculations were performed for 1, 2 and 3 opportunities, only some of these were included in IMPT’s operationalization, depending on the POI category:

- Healthcare, basic education and green spaces: 1 opportunity.
- Recreation, groceries and jobs: 3 opportunities.

3.3.2. *Mobility*

Mobility evaluates the quality of the transportation system as a service, independently of the accessibility of the destinations. The methodology to calculate its principal indicators is explained and detailed below.

3.3.2.1. *Average commuting time.*

Using the jittered (Section 3.3.1.1) OD pairs from [INE \(2018\)](#) (for commuting) and the travel time matrices, average commuting travel times are computed for each geographical unit and mode, weighted by number of trips. Trips with no feasible route within the constraints are assigned the maximum travel time (120 min) and counted separately as “not accomplishable” trips (PNA - Percentagem Não Alcançáveis).

3.3.2.2. *Average number of commuting transfers.*

For each commuting OD pair, the minimum number of transfers required for each public transport route (for 0, 1, 2, and 3 transfers) is assigned to each OD pair, and population-weighted mean transfers are computed per geographical unit. The result is the average minimum amount of transfers to commute to work using public transport.

3.3.2.3. *Public transport headways and waiting times.*

Stop-level service frequencies were computed using `tidytransit::get_stop_frequency()` ([Poletti et al., 2026](#)) for four different time windows, as can be seen in Table 5

```

headway_periods <- data.frame(
  Period = c("Weekday peak",
             "Weekday full day",
             "Weekday night",
             "Weekend"
            ),
  Day = c("Wednesday 04/02/2026", "Wednesday 04/02/2026", "Wednesday 04/02/2026", "Sunday 08/02/2026"),
  "Time window" = c("08:00-09:00", "00:00-48:00", "22:00-23:00", "10:00-11:00"),
  check.names = FALSE
)
knitr::kable(headway_periods)

```

Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
 Use 'xfun::attr2()' instead.
 See help("Deprecated")

Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
 Use 'xfun::attr2()' instead.
 See help("Deprecated")

Table 5: Public transport service frequency periods observed

Period	Day	Time window
Weekday peak	Wednesday 04/02/2026	08:00-09:00
Weekday full day	Wednesday 04/02/2026	00:00-48:00
Weekday night	Wednesday 04/02/2026	22:00-23:00
Weekend	Sunday 08/02/2026	10:00-11:00

A 500-meter buffer per stop is used to allocate stops to BGRI ([INE, 2021a](#)) population centroids. Results are aggregated to geographical units as population-weighted mean headway and waiting time. Additionally, frequency reduction factors were computed to quantify the reduction in service during night and weekend periods.

3.3.2.4. Public transport population coverage.

3.3.2.5. Ratio of pedestrian and cycling infrastructure.

The pedestrian infrastructure ratio is the proportion of road length that includes pedestrian infrastructure, computed using OpenStreetMap data:

$$R_{ped} = \frac{\text{pedestrian path length}}{\text{road length}}$$

Road classification uses the highway tags: `motorway`, `trunk`, `primary`, `secondary`, `tertiary`, `unclassified`, `residential`, `living_street`, as well as linked variants.

Pedestrian infrastructure classification uses the tags: `highway = footway` | `pedestrian` | `steps` | `residential` and `footway = sidewalk` | `crossing` | `path` | `platform` | `corridor` | `alley` | `track`, and `sidewalk = both` | `left` | `right`.

The calculation of cycling infrastructure ratio is analogous to the pedestrian ratio, and also extracted from OpenStreetMap . Cycling infrastructure is further classified according to its type and level of segregation, according to the following description and OSM tags:

- Cycle track or lane (*Ciclovía segregada*): Segregated or light-separated tracks exclusive for cycling.
 - `highway = cycleway`
 - `highway = path` & `bicycle = designated`
 - `segregated = yes` | `no`
 - `cycleway` tags containing: `separate`, `buffered_lane`, `segregated`, `lane`, or `track`.
- Advisory lane (*Via partilhada com veículos motorizados / zonas 30*): Marked cycle lanes shared with motor vehicles, such as “sharrows” or advisory lanes.
 - `cycleway` tags containing: `shared_lane`.

- Protected Active (*Via partilhada com peões*): Infrastructure shared with pedestrians but protected from motor vehicles.
 - highway = pedestrian & bicycle = designated.
 - Refinement of highway = path | footway | pedestrian when foot access is explicitly allowed on cycling paths without physical segregation.

The primary indicators are thus:

- **Cycling infrastructure ratio:**

$$R_{cyc} = \frac{\text{total cycleway length}}{\text{road length}}$$

- **Cycling quality ratio:**

$$R_{ped} = \frac{\text{segregated cycleway length}}{\text{total cycleway length}}$$

Where “segregated cycleway length” corresponds to the “cycle track or lane” category.

3.3.2.6. Shared mobility availability.

The number of shared mobility points (physical and virtual docks from all operators in the LMA ([TML and Way2Go, 2025](#))) was counted and aggregated per parish and per grid cell.

3.3.3. Affordability

Affordability evaluates the capacity of households to support the monetary costs of transportation to access essential activities. IMPT uses a weighted cost by mode, in which the monetary cost of automobile/public transport travel is multiplied by each parish’s modal share, and expressed as a proportion of median household income. The travel costs are estimated for one-way trips from home to the workplace during the morning peak.

3.3.3.1. Cost of driving.

The cost of car travel is computed through:

$$Cost_{car} = d \times 0.40\text{€/km} + Cost_{toll}$$

Where a fixed cost of 0.40€/km is used, based on [República Portuguesa \(2008\)](#), which sets mileage allowance to cover fuel and car maintenance. Tolls are computed through the toll calculator of Infraestruturas de Portugal ([IP, 2026](#)), using the route geometry returned by `r5r::detailed_itineraries()`.

3.3.3.2. Cost of public transportation.

To account for uncertainty in public transport fare payments, two scenarios were considered:

Scenario A - Single ticket

The average single ticket cost was calculated based on revenue data from Carris ([Carris, 2024](#)):

- Total single ticket validations: 17,424,000
- Total single ticket revenue: 33,278,000€
- Average cost per single ticket: ~1.91€

This way, a single ticket cost of **1.91€** was assumed for all modes. Free transfers within the same mode for 60 minutes are assumed, but transfers to different modes incur an additional charge.

Scenario B - Monthly pass (Navegante)

Based on operator revenue data, the average cost of a monthly pass was estimated (Table 6).

```
navegante <- data.frame(
  "Pass type" = c("Navegante Metropolitano",
                  "Navegante Municipal",
                  "Senior (+65)",
                  "Family",
                  "Youth (sub-23)"
                ),
  "Pass recharges" = c("5,063,510", "887,190", "1,103,782", "243,784", "3,425,013"),
  "Pass price" = c("40€", "30€", "20€", "80€", "0€"),
  check.names = FALSE
)
knitr::kable(navegante)
```

Warning in attr(x, "align"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Warning in attr(x, "format"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Table 6: Revenue from Navegante monthly pass

Pass type	Pass recharges	Pass price
Navegante Metropolitano	5,063,510	40€
Navegante Municipal	887,190	30€
Senior (+65)	1,103,782	20€
Family	243,784	80€
Youth (sub-23)	3,425,013	0€

The average cost paid by a monthly pass holder is thus ~**25.25€**. Assuming an average of 44 monthly trips (2 trips in each of 22 working days), the average cost per trip becomes

~0.57€.

3.3.3.3. Composite indicators.

Transport expenditure is expressed as a share of household income, with two variants (with and without housing costs). All costs are annualized and computed as yearly expenses per household (€/year/household). 250 days per year were considered, with two trips per day. Tolls are considered assymetrically for both directions, since the bridges' tolls are only paid one-way.

For driving:

$$transp_inc_{car} = \frac{C_{car}}{I_{household}}$$

$$h_transp_inc_{car} = \frac{C_{car} + C_{housing}}{I_{household}}$$

To combine car and public transport costs, a composite indicator based on each parish's modal share from [INE \(2021b\)](#) is computed:

$$transp_inc_{weighted} = \frac{C_{car} \cdot s_{car} + C_{PT} \cdot s_{PT}}{s_{car} + s_{PT} + s_{active}}$$

Where s_m is the modal share of mode m . Active modes (walking and cycling) are included in the denominator to preserve the real modal structure of each parish, but are not included in the numerator, given their null direct cost. This option avoids overestimating the financial burden in parishes with a high active modal share, while reflecting the fact that a part of the population travels without a cost.

Driving costs are further adjusted for vehicle occupancy rate, household size and mobile population fraction ([INE, 2018](#)), to convert per-trip costs to a household-level annual burden. This way, the final average annual transport expenditure, accounting for all adjustments and modal shares, is computed as:

$$transp_inc_comp = \frac{HH_{size} \cdot P_{mobile}}{I_{household}} \cdot \left[\left(\frac{C_{car,day}}{O} \cdot \frac{s_{car}}{s_{car} + s_{PT} + s_{active}} \right) + \left(C_{PT,day} \cdot \frac{s_{PT}}{s_{car} + s_{PT} + s_{active}} \right) \right] \cdot N_d$$

Where:

- N_{days} = 250 working days per year.
- HH_{size} is the average household size.
- P_{mobile} is the share of mobile population.
- $I_{household}$ is the average annual household income.
- $C_{car,day}$ and $C_{PT,day}$ are the round-trip costs for car and public transport.
- O is the average vehicle occupancy rate.
- s_{car} , s_{PT} and s_{active} are the modal shares (INE, 2021b).

For the variant that includes housing costs ($h_transp_inc_comp$), the annualized housing cost $C_{housing,year}$ (INE, 2022) is added to the numerator.

3.3.4. Safety

This dimension encompasses the disproportional exposure to externalities of the transportation system. It is entirely based on road safety data provided by Autoridade Nacional de Segurança Rodoviária (ASNR) through Transportes Metropolitanos de Lisboa (TML) (TML and Way2Go, 2025). It is important to note that only road crashes occurring *inside localities* (dentro das localidades), a category specified by the ASNR itself, are included in the index. In particular, this excludes crashes on motorways and on the two bridges crossing the Tagus river, while maintaining IMPT's spatial scope.

The indicators calculated for each geographical unit and transport mode are systematized below:

- **Total accidents (5 years):**

$$\sum_{25} (Acc_{5y})$$

- **Fatalities (30-day):**

$$\sum(VM_{30d})$$

- **Serious injuries (30-day):**

$$\sum(FG_{30d})$$

- **Slight injuries (30-day):**

$$\sum(FL_{30d})$$

- **Accident rate:**

$$\frac{\sum(Acc_{5y})}{Population} \times 1000$$

- **Fatality rate:**

$$\frac{\sum(VM_{30d})}{Population} \times 1000$$

- **Severity index (total):**

$$\frac{\sum(VM_{30d})}{\sum(Acc_{5y})}$$

- **Severity index (motorized):**

$$\frac{\sum(VM_{30d})}{Motorized\ vehicles\ involved}$$

- **Severity index (cycling):**

$$\frac{\sum(VM_{30d})}{Cyclists\ involved}$$

- **Severity index (pedestrian):**

$$\frac{\sum(VM_{30d})}{Pedestrians\ involved}$$

3.4. Normalization and Aggregation

All indicators were normalized to the range [1, 100] using min-max scaling:

For benefit indicators (higher raw value $\rightarrow$ better):

$$X_{norm} = 1 + \frac{X - X_{min}}{X_{max} - X_{min}} \times 99$$

For cost indicators (lower raw value $\rightarrow$ better):

$$X_{norm} = 1 + \frac{X_{max} - X}{X_{max} - X_{min}} \times 99$$

- **Accessibility** metrics (`access_*`) are *benefit* indicators.
- **Mobility** cost metrics (`mobility_cost_*`), public transport waiting times, all **Affordability** indicators, and all **Safety** indicators are *cost* indicators.

Missing values in accessibility indicators (arising from unfeasible public transportation routes) are imputed with the mean of the respective variable prior to PCA.

3.5. PCA-Based Dimension Scores

For every dimension with multiple indicators (Accessibility, Mobility and Safety) and at the global (all-mode) and per-mode levels, a PCA is run using `FactoMineR::PCA()` at parish level.

In line with [OECD \(2008\)](#), the first principal component (PC1) score is used as the dimension index after rescaling to [1, 100]. PC1 captures the dominant axis of variation among the normalised indicators.

For the Affordability dimension, a PCA is not applied, due to there being too few indicators. Instead, the normalized composite `h_transp_inc_comp` (housing + transport cost as share of income, modal-share weighted) is used directly as the Affordability Index.

For Safety per mode, an equal-weight average of the severity index (per mode) and total vehicles involved (per mode) is used instead of PCA.

For Mobility - Car, a PCA is not applied, also due to there being too few indicators. Instead, we assumed an equal-weight average of the normalized composite `mobility_commuting_avg_tt_car` and `road_length`.

3.6. Composite IMPT Scores

Three aggregation methods are computed for the global IMPT and each per-mode IMPT:

1. **Critical IMPT - geometric mean** (equal weights, Figure 3a):

$$IMPT_{geom} = \left(\prod_d (S_d + 1) \right)^{1/n} - 1$$

Partially compensatory: penalizes dimensions with very low values non-linearly. Highlights the most critical situations, where at least one dimension is severely deficient.

2. **Balanced IMPT - arithmetic mean** (equal weights, Figure 3b):

$$IMPT_{avg} = \frac{1}{n} \sum_d S_d$$

Compensatory: high vulnerability in one dimension can be offset by good performance in others. Suitable for setting aggregate policy targets and communicating to non-specialist audiences.

3. **Contrasting IMPT - entropy-weighted geometric mean** (Figure 3c):

$$IMPT_{entropy} = \prod_d (S_d + 1)^{w_d} - 1$$

where weights w_d are derived from the **entropy weighting method**:

Weights are determined by the empirical dispersion of the dimensions across parishes: the dimension that varies most among territories receives greater weight. Maximizes territorial contrast and is useful for competitive prioritization of interventions.

$$w_d = \frac{d_d}{\sum_j d_j}, \quad d_d = 1 - e_d, \quad e_d = -k \sum_i p_{id} \ln(p_{id})$$

with $k = 1/\ln(n)$ and p_{id} being the proportion of the dimension score for area i .

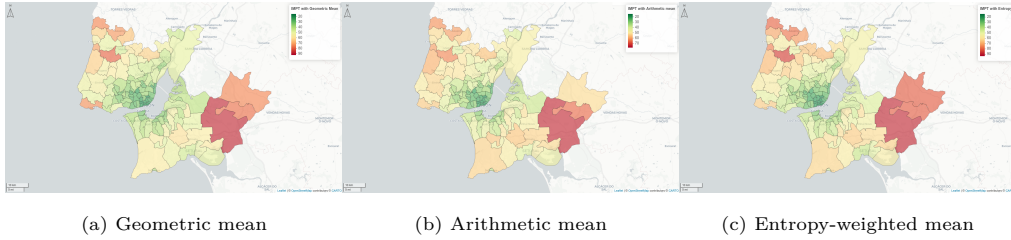


Figure 3: IMPT results for different aggregation methods for all modes (with Navegante)

3.7. Validation

Finally, a workshop was carried out with LMA municipalities. It took place on April 9th, 2026 in the offices of Transportes Metropolitanos de Lisboa (TML), with the objectives of not only presenting and validating the results with municipality technicians, but also confronting the computed data with the local perception of the participants, in order to identify relevant divergences. It included representatives from 13 out of the 18 municipalities in the LMA, as well as representatives from transport authorities and operators.

4. Results

The full results of the IMPT for all LMA parishes are available in the dashboard created in the scope of this project. This dashboard allows for the visualization of the global indices (critical, balanced and contrasting), the results of each dimension by transport mode, all indicators that make up the index, as well as key statistics and summaries for various scales (parish, municipality and grid). Beyond results visualization, this tool allows for the calculation of the aggregated index adjusted to user preferences (political IMPT). For this purpose, an Analytic Hierarchical Process (AHP), that compares the user's preferences between dimensions and this way assigns weights to each, is made available to the dashboard's users. The dashboard is available at: <https://ushift.tecnico.ulisboa.pt/apps/impt>.

It is worth noting that the IMPT is a relative index, created with the purpose of comparing different areas in the same metropolitan area, not of being an absolute index of transport poverty. This way, a global score of 100 or 0 should not be interpreted as the “best” or “worst” possible outcomes.

4.1. Methodological comparison

4.2. Spatial performance and outliers

4.3. Takeaways

Figure 4 is generated using an R chunk.

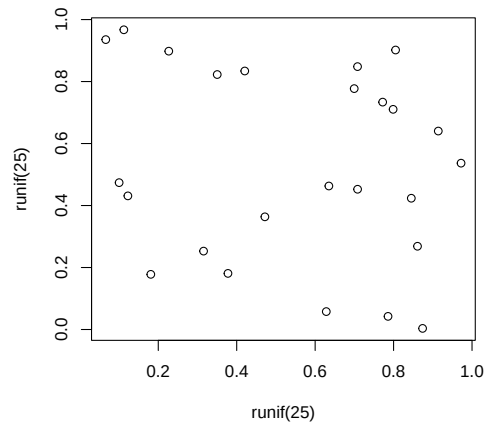


Figure 4: A meaningless scatterplot

5. Discussion

5.1. Limitations

Tables can also be generated using R chunks, as shown in Table 7 example.

```
knitr::kable(head(mtcars)[,1:4])
```

Warning in attr(x, "align"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Warning in attr(x, "format"): 'xfun::attr()' is deprecated.

Use 'xfun::attr2()' instead.

See help("Deprecated")

Table 7: Caption centered above table

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93

Table 7: Caption centered above table

	mpg	cyl	disp	hp
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

6. Conclusions

This is the best paper ever.

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